

Unit 12: Prediction and Classification Using Weka Tool

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Objective

After this lecture, you will be able to

- How to use Weka to create a decision tree model.
- Understand the creation and working of Naïve Bayes in weka.
- Know the various applications of web mining.

Introduction

Weka is a set of data mining-related machine learning techniques. It includes data preparation, categorization, regression, clustering, mining of association rules, and visualization tools. It provides access to a vast variety of classification algorithms. One of the advantages of using the Weka platform to solve your machine learning challenges is the wide variety of machine learning algorithms available. Weka comes with many built-in functions for constructing a variety of machine learning techniques, ranging from linear regression to neural networks. With just a press of a button, you can install even the most complex algorithms on your dataset! Not only that, but Weka also allows you to use Python and R to access some of the most popular machine learning library algorithms.

12.1 Naïve Bayes Using Weka

The Bayes' Theorem is used to create the Naive Bayes classifiers, which are a series of classification algorithms based on the Bayes' Theorem. It's a group of algorithms that all have the same premise: each pair of features to be classified is independent of the others. This experiment's "weather-nominal" data set is available in ARFF format.

Steps to be followed

1. Using the choose file option, we must first load the necessary dataset into the Weka tool. We're going to run the weather-nominal dataset here.

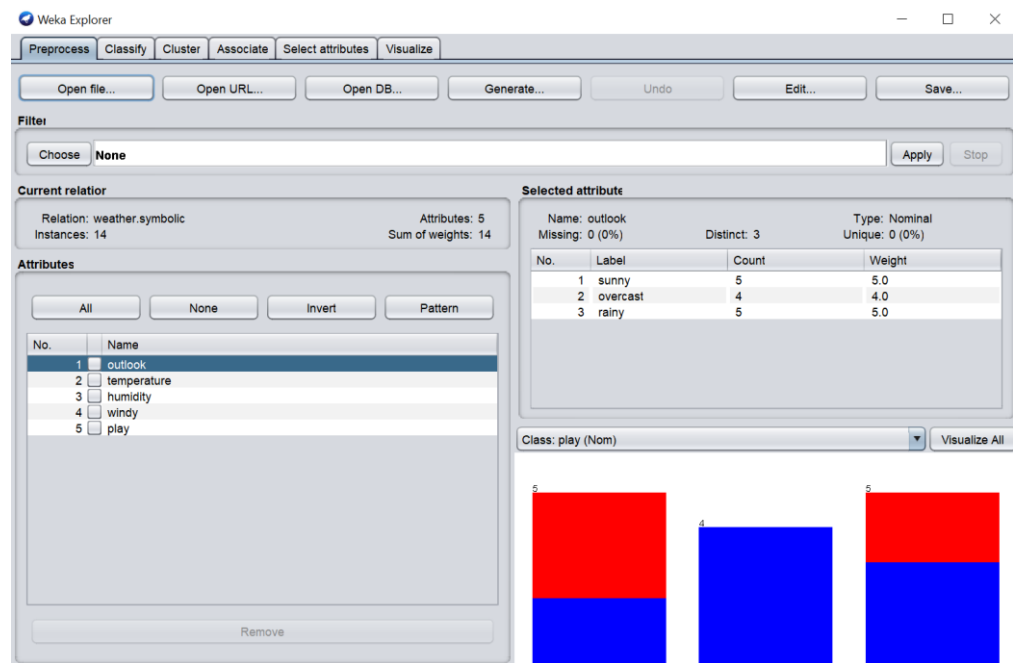


Figure 1: Dataset Selection

- Now, on the top left side, go to the classify tab, click the choose button, and pick the Naive Bayesian algorithm.

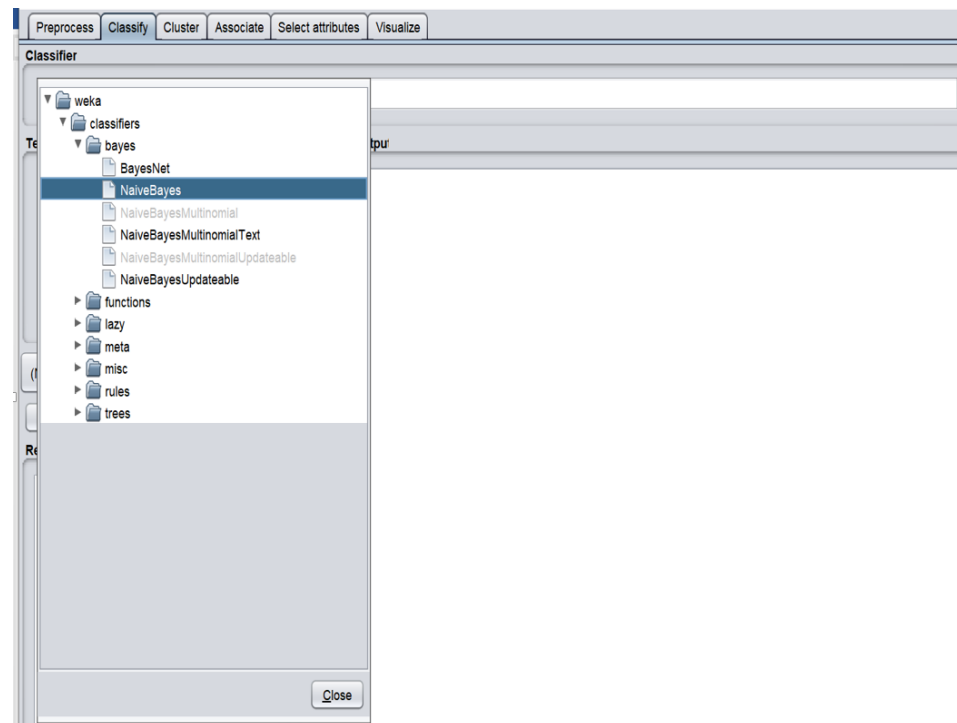


Figure 2: Selection of Naive Bayes

- To adjust the parameters, click the choose button on the right side, and in this example, we'll accept the default values.

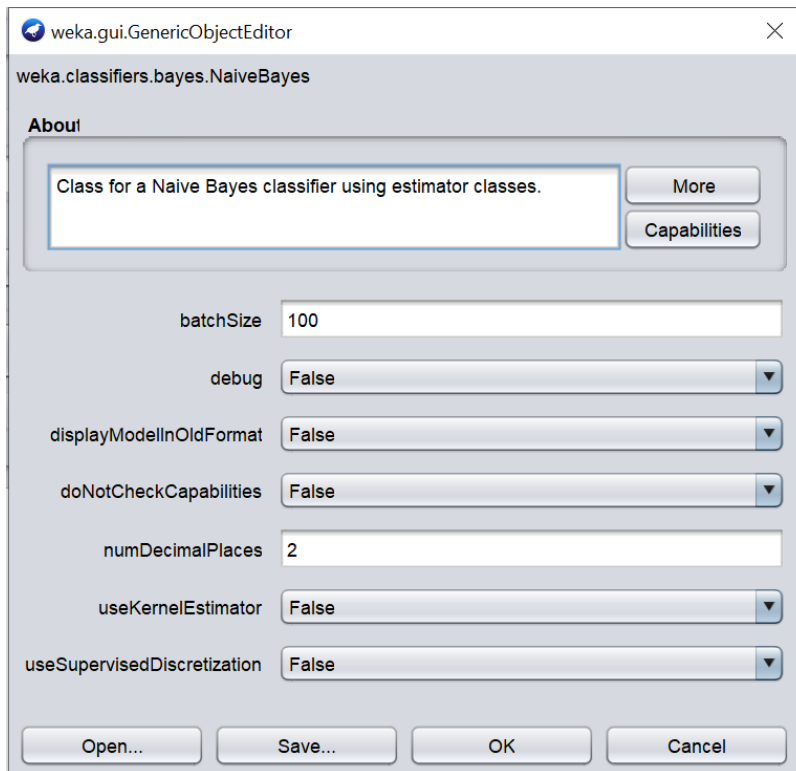


Figure 3: Adjustment of parameters

4. From the "Test" options in the main panel, we choose Percentage split as our measurement process. We'll use the percentage split of 66 percent to get a clear estimate of the model's accuracy since we don't have a separate test data set. There are 14 examples in our dataset, with 9 being used for training and 5 for testing.

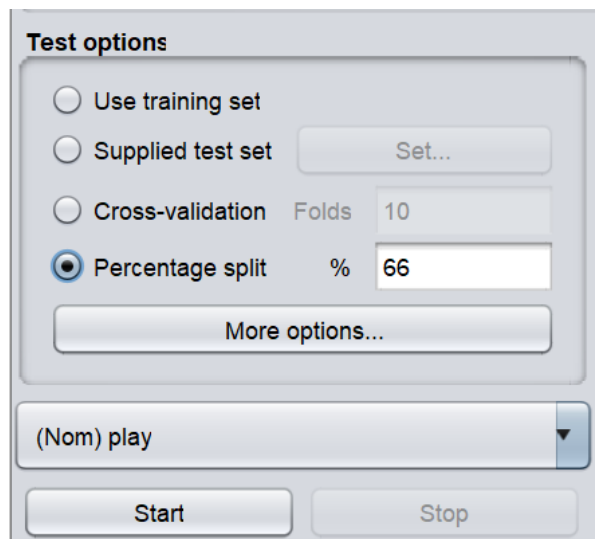


Figure 4: Selection of Split Ratio

5. We'll now click "start" to begin creating the model. The assessment statistic will appear in the right panel after the model has been completed.

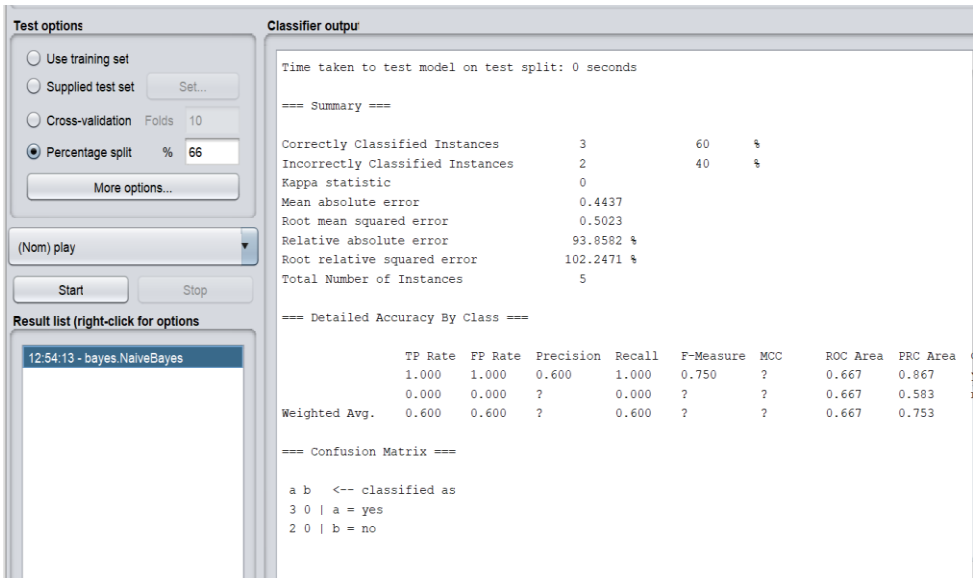


Figure 5: Model Output

6. It's worth noting that the classification accuracy of the model is about 60%. This means that by making certain changes, we would be able to improve the accuracy. (Either during preprocessing or when choosing current classification parameters.)

Furthermore, we can use our models to find new instances. In the main panel's "Test options," select the "supplied test package" radio button, then click the "set" button.

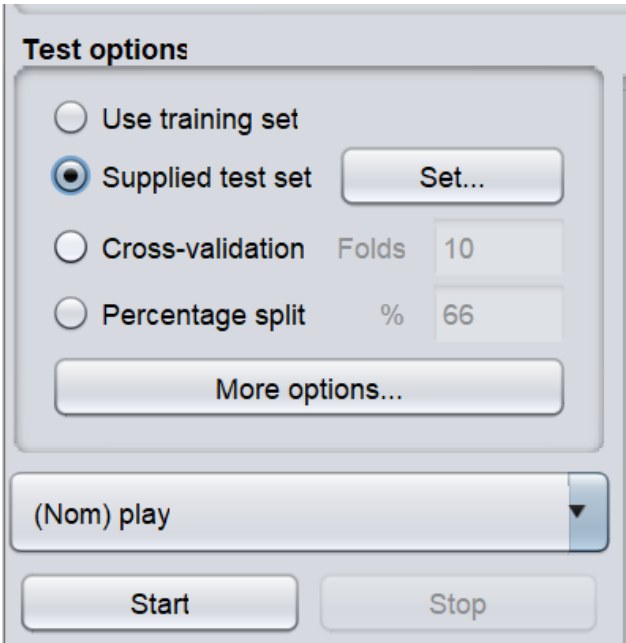


Figure 6: Selection of supplied Test set

This will open a pop-up window that will allow us to open the test instance file. It can be used to further increase the accuracy of the module by using different test sets.

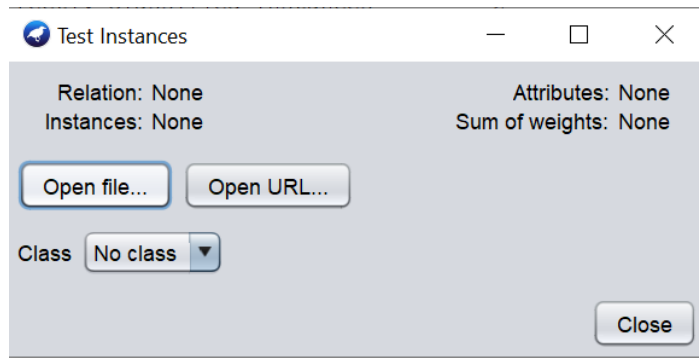


Figure 7: Selection of test Instances



Using the *airline.arff* inbuilt file from Weka dataset and implement Naïve Bayes algorithm on it.

12.2 Classification using Decision Tree in Weka

Classifier J48

It is a C4.5-generated algorithm for generating a decision tree (an extension of ID3). A statistical classifier is another name for it. A database is needed for decision tree classification.



Certain algorithms are greyed out depending on whether the problem is a classification or regression problem (piecewise linear function M5P for regression). Besides that, certain decision trees have distinct Weka implementations (for example, J48 implements C4.5 in Java)

In Weka, creating a decision tree is relatively easy. Simply follow the steps below:

1. Open WEKA explorer.
2. Choose the *weather.nominal.arff* file from the preprocess tab's "choose file" option.

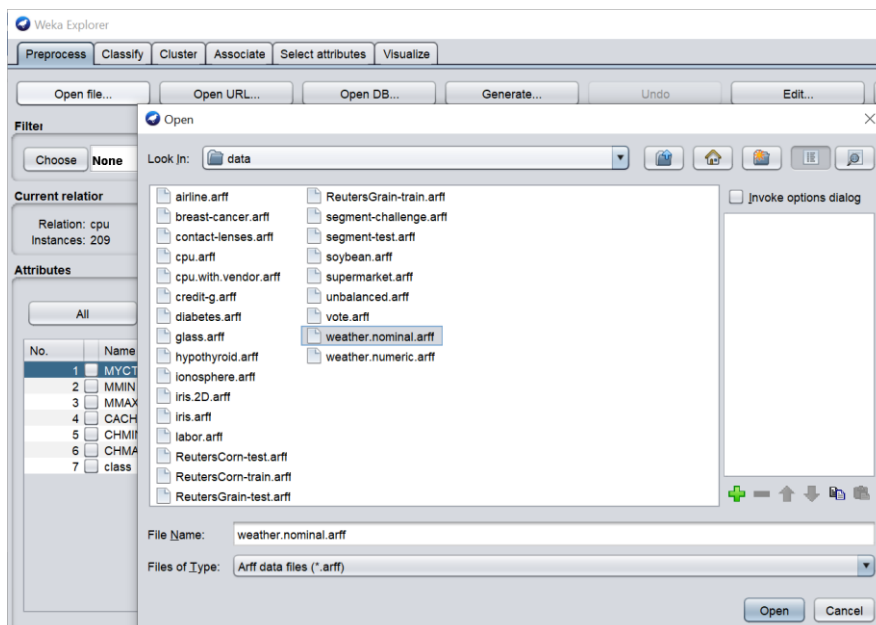


Figure 8: Selection of Dataset

3. To classify the unclassified data, go to the "Classify" tab. Select "Choose" from the drop-down menu. Select "trees -> J48" from this menu.

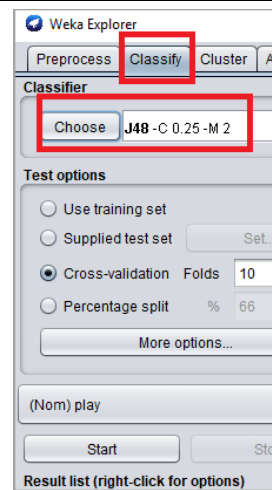


Figure 9: Selection of Decision tree(J48)

4. To begin, press the Start Button. The performance of the classifier will be visible on the right-hand panel. The run information is shown in the panel as follows:

Scheme: The classification algorithm that was used is referred to as the scheme.

Instances: Number of data rows in the dataset.

Attributes: There are five attributes in the dataset.

The decision tree is defined by the number of leaves and the size of the tree.

Time taken to build the model: Time for the output.

Full classification of the J48 pruned with the attributes and number of instances.



“A decision tree divides nodes based on all available variables and then chooses the split that produces the most homogeneous sub-nodes.” The homogeneity of a sample at a split is calculated using Information Gain.

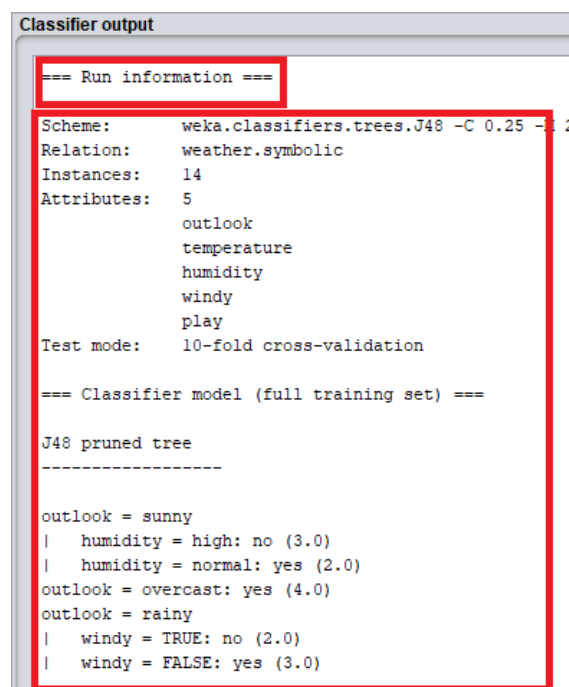


Figure 10: J48 output for the selected dataset

Right-click on the result and choose to visualize the tree from the list.

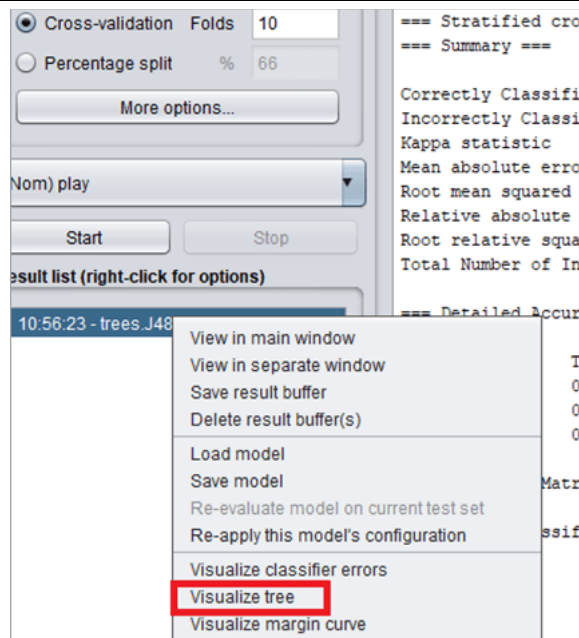


Figure 11: Selection of visualizing tree

5. The output is in the form of a decision tree. The main attribute is “outlook”.

If the outlook is sunny, then the tree further analyzes the humidity. If humidity is high then the class label play= “yes”.

If the outlook is overcast, the class label, play is “yes”. The number of instances which obey the classification is 4.

If the outlook is rainy, further classification takes place to analyze the attribute “windy”. If windy=true, the play = “no”. The number of instances which obey the classification for outlook= windy and windy=true is 2.

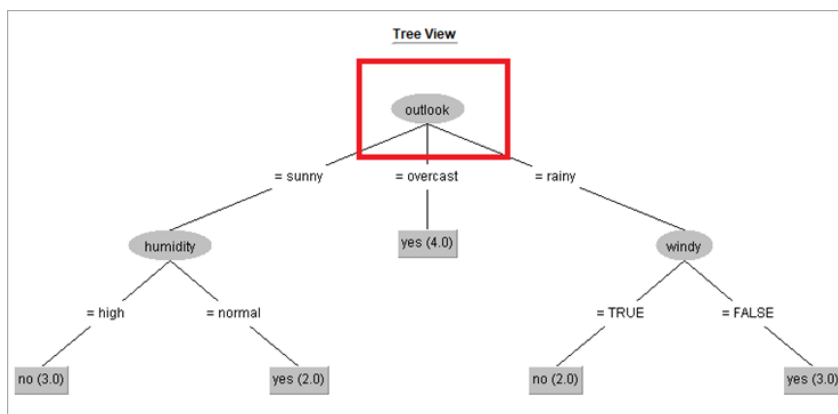


Figure 12: Tree view of J48



How would this instance be classified using the decision tree? outlook = sunny, temperature = cool, humidity = high, windy = TRUE.

12.3 Applications of classification for web mining

Web mining is the process of extracting knowledge from web data, such as web content, web structure, and web usage data, using data mining techniques. Web mining's major goal is to extract relevant information from the World Wide Web and its user trends. Web mining is classified into three categories: web content mining, web structure mining, and web usage mining. These are explained in the following sections.

Web Content Mining

The application of obtaining relevant information from the content of web documents is known as web content mining. Text, image, audio, video, and other sorts of data form web content.

A web page's content data is a collection of information. It has the potential to provide useful and interesting patterns about user requirements. Text mining, machine learning, and natural language processing are all connected to text documents. Text mining is another name for this type of mining. According to the content of the input, this sort of mining scans and mines text, images, and groups of web pages. It can be further classified into:

- **Web page content mining:** The typical search of a web page by content is known as content mining.
- **Search result mining:** The term "search result mining" refers to the second search of sites found in a prior search.

In web content mining, there are two approaches:

1. Agent-based approach
2. Database approach

In the Agent-based approach, there are three types of agents

- Intelligent search agents
- Information filtering/Categorizing agent
- Personalized web agents.

Intelligent search agents: Intelligent Search agents automatically searches for information according to a particular query using domain characteristics and user profiles.

Information filtering/Categorizing agent: To filter data according to predefined instructions, information agents used a variety of techniques.

Personalized web agents: Personalized web agents learn about user preferences and find documents that are relevant to those profiles.

It comprises a well-formed database comprising schemas and attributes with defined domains in the Database approach.



If a user wants to search for a particular book, then the search engine provides a list of suggestions.

Web structure mining

The application of discovering structural information from the web is known as web structure mining. The web graph's structure is made up of nodes (web pages) and edges (links) that connect related pages. Structure mining essentially displays a structured summary of a webpage. It establishes a link between web pages that are linked by information or a direct link. To determine the connection between two commercial websites, Web structure mining can be very useful.

Web structure mining, one of three types of data mining, is a technique for determining the relationship between Web pages linked by information or direct links. It provides details on how various pages are connected to construct this massive web.

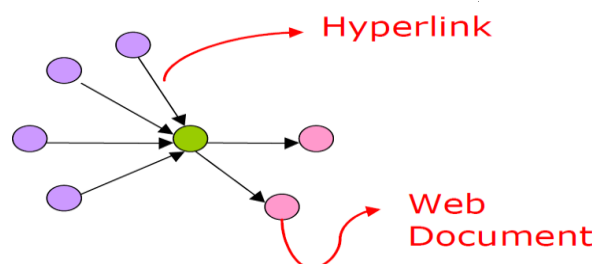


Figure 13: Connection between webpages

Web pages serve as nodes in a typical Web graph, while hyperlinks serve as edges linking two related pages. Web Structural Mining is the method of extracting structure data from the internet. This type of mining can be done at the document (intra-page) level or the hyperlink (inter-page) level. Hyperlink Analysis is a term used to describe study at the level of hyperlinks.

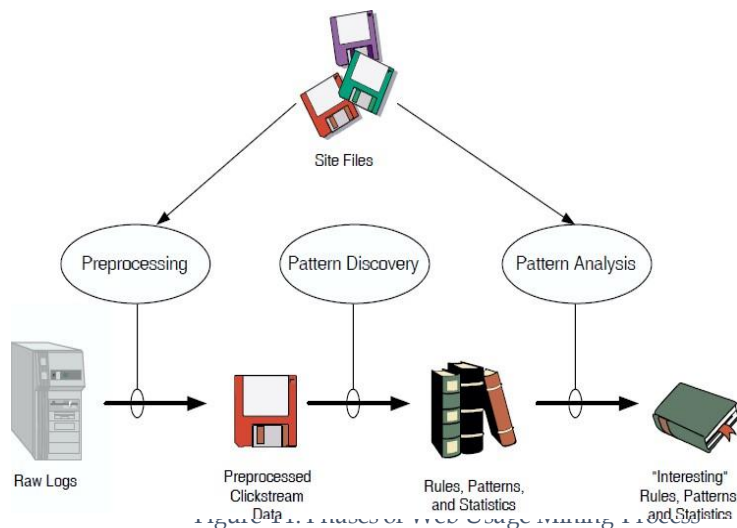


Web structure mining can be very useful to companies to determine the connection between two commercial websites.

Web Usage Mining

Web usage mining, a subset of data mining, is the extraction of various types of interesting data that is readily available and accessible on the Internet—or formally known as the World Wide Web—in the ocean of massive web pages (WWW). It has aided in the analysis of user actions on various web pages and tracking them over time as one of the uses of data mining methodology. Web usage mining may reveal relationships that were not intended by the pages' creators.

Some of the methods to identify and analyze the web usage patterns are given below:



Preprocessing: Preprocessing entails translating the following information from diverse data sources:

- usage information
- content information
- structure information

into the data abstractions required for pattern identification from the many available data sources.

Pattern Discovery: Pattern recognition uses methods and algorithms from a variety of domains, including statistics, data mining, machine learning, and pattern recognition.

Pattern Analysis: Pattern analysis is the final step in the whole Web Usage mining process. Its goal is to eliminate uninteresting rules or patterns from the set discovered during the pattern discovery phase. The application for which Web mining is done usually determines the exact analytic methodology. The most common form of pattern analysis consists of:

- A knowledge query technique, such as SQL, is the most prevalent kind of pattern analysis.
- Another method is to input user data into a data cube and use it for Online Analytical Processing (OLAP).
- Visualization techniques like graphing patterns or assigning colors to distinct values can often reveal broad patterns or trends in data.
- Patterns having pages of a specific usage kind, content type, or pages that fit a specific hyperlink structure can be filtered out using content and structure information.

Challenges in Web Mining

Based on the following observations, the web pretends to present incredible challenges for resource discovery and knowledge discovery:

The complexity of Web Pages

There is no unified structure to the web pages. When compared to regular text documents, they are exceedingly complex. In the web's digital library, there are tremendous volumes of documents. These libraries are not arranged in any particular sequence.

The web is a Dynamic Data Source

The information on the internet is constantly updated. News, weather, shopping, financial news, sports, and so on are only a few examples.

Client networks are diverse

The web's client network is rapidly growing. These customers have a variety of hobbies, backgrounds, and purposes for using the service. There are over a hundred million internet-connected workstations, and this number is rapidly growing.

Relevance of Data

It is assumed that a specific person is only concerned with a limited area of the internet, while the rest of the web contains data that is unfamiliar to the user and may lead to unexpected results.

Summary

- Naive Bayes is a classification algorithm. Traditionally it assumes that the input values are nominal, although numerical inputs are supported by assuming a distribution.
- Naïve Bayes classifier is a statistical classifier. It assumes that the values of attributes in the classes are independent.
- J48 can deal with both nominal and numeric attributes
- Decision trees are also known as Classification And Regression Trees (CART).
- Each node in the tree represents a question derived from the features present in your dataset.
- Weka is free open-source software that comes with several machine learning algorithms that can be accessed via a graphical user interface.
- Pattern analysis is the final step in the whole Web Usage mining process.
- Naïve Bayes classifiers assume that the effect of a variable value on a given class is independent of the values of other variables. This assumption is called class conditional independence. It is made to simplify the computation and in this sense considered to be "Naïve".

Keywords

Bayesian classification: Bayesian classifiers are statistical classifiers.

Classification: Classification is a data mining technique used to predict group membership for data instances.

Decision Tree: A decision tree is a flow-chart-like tree structure, where each *internal node* denotes a test on an attribute, each *branch* represents an outcome of the test, and *leaf nodes* represent classes or class distributions. The topmost node in a tree is the *root* node.

Predictive accuracy: This refers to the ability of the model to correctly predict the class label of new or previously unseen data.

Intelligent search agents: Intelligent Search agents automatically searches for information according to a particular query using domain characteristics and user profiles.

Web page content mining: The typical search of a web page by content is known as content mining.

Instances: Number of data rows in the dataset.

Self Assessment Questions

1. In WEKA explorer _____ tab we need to open data file.
 - A. Preprocess
 - B. Cluster
 - C. Select Attribute
 - D. Classify
2. Which of the following tab is used to build a Naive Bayes model.
 - A. Preprocess
 - B. Cluster
 - C. Select Attribute
 - D. Classify
3. To build a classifier in Weka we need to click on
 - A. Build Button
 - B. Start Button
 - C. Create Button
 - D. None
4. Under classify you need to select _____ button, and from the tree menu, select Naïve Bayes.
 - A. Choose Classifier
 - B. Select Classifier
 - C. Create Classifier
 - D. Build Classifier
5. In Weka which of the following options is/are available under test options.
 - A. Use Training Set
 - B. Supplied Test Set
 - C. Cross-Validation
 - D. All of the above.
6. _____ work by learning answers to a hierarchy of if/else questions leading to a decision.
 - A. Decision trees
 - B. KNN
 - C. Naïve Bayes
 - D. Perceptron
7. Which of the following steps are used to load the dataset in Weka.
 - A. Open Weka GUI
 - B. Select the “Explorer” option.
 - C. Select “Open file” and choose your dataset.
 - D. All of the above
8. Implementing a decision tree in Weka from the drop-down list, select _____ which will open all the tree algorithms.
 - A. Tree
 - B. Trees
 - C. Decision Tree

D. ALL Tree

9. In decision tree _____ is used to calculate the homogeneity of the sample at a split.

A. Information Gain

B. No of attributes

C. Entropy

D. Kappa Statistics

10. Decision tree splits the nodes on all available variables and then selects the split which results in the most _____ sub-nodes.

A. Homogeneous

B. Heterogeneous

C. Both

D. None

11. For what purpose, the analysis tools pre-compute the summaries of the huge amount of data?

A. To maintain consistency

B. For authentication

C. For data access

D. To obtain the queries response

12. Which of the following statement is true about the classification?

A. It is a measure of accuracy

B. It is a subdivision of a set

C. It is the task of assigning a classification

D. None of the above

13. Task of inferring a model from training labeled data is called

A. Unsupervised Learning

B. Supervised Learning

C. Reinforcement Learning

D. All of the above

14. _____ are used in data mining to classify data based on class labels.

A. Classification algorithms

B. Clustering Algorithms

C. Both

D. None

15. _____ is the process of Data Mining techniques to automatically discover and extract information from Web documents and services.

A. Text Mining

B. Data Mining

C. Web Mining

D. None

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. B | 4. A | 5. D |
| 6. A | 7. D | 8. B | 9. A | 10. A |
| 11. D | 12. B | 13. B | 14. A | 15. C |

Review Questions

- Q1. Create a dataset and apply a decision tree algorithm on it and display the results in the tree form.
- Q2. Consider training and predicting with a naive Bayes classifier for two document classes. The word “booyah” appears once for class 1, and never for class 0. When predicting new data, if the classifier sees “booyah”, what is the posterior probability of class 1?
- Q3. Elucidate the concept of web mining along with its various categories.
- Q4. With the help of example explain the various challenges of Web mining.
- Q5. Consider the weather.arff file and discuss why does it make more sense to test the feature “outlook” first? Implement the dataset using decision tree.

Further Reading

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